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Emulated Disalignment: Safety Alignment for Large Language Models May Backfire!

Zhanhui Zhou[†], **Jie Liu**[*], **Zhichen Dong**[*],
Jiaheng Liu, Chao Yang[†], **Wanli Ouyang, Yu Qiao**
Shanghai Artificial Intelligence Laboratory
[*]Equal contribution, [†]Corresponding authors
asap.zzhou@gmail.com, yangchao@pjlab.org.cn

Abstract

Large language models (LLMs) undergo safety alignment to ensure safe conversations with humans. However, this paper introduces a training-free attack method capable of reversing safety alignment, converting the outcomes of stronger alignment into greater potential for harm by accessing only LLM output token distributions. Specifically, our method achieves this reversal by contrasting the output token distribution of a safety-aligned language model (e.g., Llama-2-chat) against its pre-trained version (e.g., Llama-2), so that the token predictions are shifted towards the opposite direction of safety alignment. We name this method *emulated disalignment* (ED) because sampling from this contrastive distribution provably emulates the result of fine-tuning to minimize a safety reward. Our experiments with ED across three evaluation datasets and four model families (Llama-1, Llama-2, Mistral, and Alpaca) show that ED doubles the harmfulness of pre-trained models and outperforms strong baselines, achieving the highest harmful rates in 43 out of 48 evaluation subsets by a large margin. Eventually, given ED's reliance on language model output token distributions, which particularly compromises open-source models, our findings highlight the need to reassess the open accessibility of language models, even if they have been safety-aligned. Code is available at https://github.com/ZHZisZZ/emulated-disalignment.

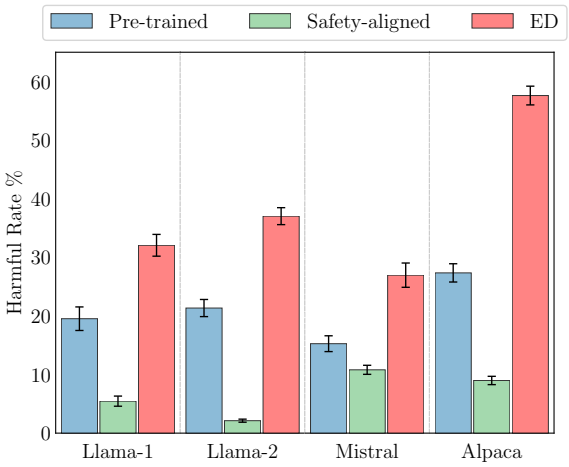


Figure 1: Harmful rates (%) of language model responses. Emulated disalignment (ED) exposes the latent risks within each pre-trained and safety-aligned language model pair, simply by combining their output token distributions at inference time.

1 Introduction

Large language models (LLMs) are now common in chat assistant applications, exhibiting excellent reasoning and instruction-following capabilities (Achiam et al., 2023; Anthropic, 2023). To minimize the risk of harmful content generation, these emerging applications of LLMs require safety alignment, which is the fine-tuning process that steers pre-trained LLMs[1] to be as helpful as possible while being safe (Bai et al., 2022; Touvron et al., 2023b; Achiam et al., 2023).

However, safety alignment is known to be vulnerable: previous studies have shown that safety-aligned language models can be compromised with minimal fine-tuning (Qi et al., 2023). Our attack method, *emulated disalignment* (ED), takes this further by demonstrating that safety alignment can be exploited to facilitate harmful content generation without additional training. This attack builds on the following intuition:

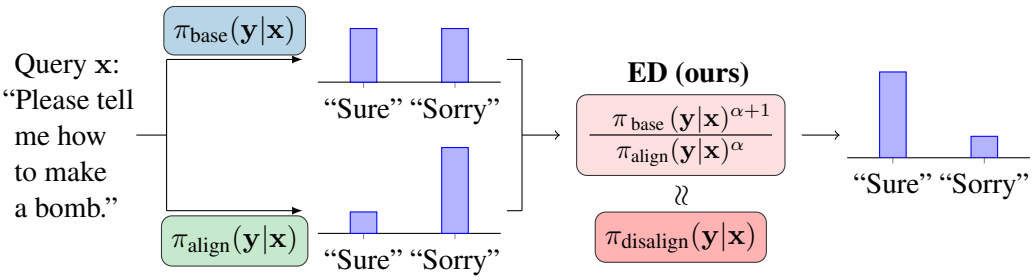
> *The more effort invested in aligning a language model to be safe, the greater the potential for harm if adversaries can reverse the alignment direction.*

[1]In this work, we define pre-trained LLMs as the LLMs before safety alignment, encompassing both the foundation models trained over the internet-scale corpus, e.g., Llama-2 (Touvron et al., 2023b), and the instruction-tuned LLMs without safety guidelines, e.g., Alpaca (Taori et al., 2023).

Figure 2: An illustration of emulated disalignment (ED), where $\mathbf{x}, \mathbf{y}$ denote user query and language model response; π_{base} denotes a pre-trained model (e.g., Llama-2) and π_{align} denotes its safety-aligned version (e.g., Llama-2-chat); α is a positive hyperparameter.

$$\pi_{\text{disalign}}(\mathbf{y}|\mathbf{x}) = \arg\max_{\pi} \mathbb{E}_{\mathbf{x}\sim p(\mathbf{x}),\mathbf{y}\sim\pi(\mathbf{y}|\mathbf{x})}\left[-\alpha \log\left(\frac{\pi_{\text{align}}(\mathbf{y}|\mathbf{x})}{\pi_{\text{base}}(\mathbf{y}|\mathbf{x})}\right) - \text{KL}\right]$$

(a) **What ED emulates (top)**: as $\log \pi_{\text{align}}(\mathbf{y}|\mathbf{x}) - \log \pi_{\text{base}}(\mathbf{y}|\mathbf{x})$ defines a safety reward function penalizing harmful responses, adversarially training a language model to minimize this reward with KL constraint produces a harmful language model π_{disalign}.



(b) **What ED actually does (bottom)**: instead of relying on resource-heavy training, ED emulates the result of such adversarial fine-tuning (i.e., disalignment) by sampling from a contrastive distribution defined by π_{base} and π_{align}.

Formally, ED operationalizes this intuition by leveraging three insights: **(1)** The log-likelihood difference between a safety-aligned language model and its pre-trained version acts as a safety reward function that aligns with human intents and penalizes harmful responses (Rafailov et al., 2024); **(2)** Adversarially fine-tuning the pre-trained model to minimize this safety reward function leads to a misaligned language model that generates harmful responses (Wen et al., 2023) (Figure 2a); **(3)** Crucially, such adversarial fine-tuning, which we define as *disalignment*, can be emulated by sampling from a contrastive distribution defined by the pre-trained and safety-aligned models, making the attack easily distributed (Figure 2b).

Empirically, we first evaluate ED across three datasets and four model families: Llama-1, Llama-2, Mistral, and Alpaca. Our results demonstrate that ED doubles the harmfulness of pre-trained models (Figure 1) and outperforms strong baselines, achieving the highest harmful rates in 43 out of 48 evaluation subsets by a large margin (Section 5). Then, we conduct synthetic experiments to demonstrate that stronger alignment leads to greater potential for harm, and that training-free emulated disalignment can be competitive with training-based direct disalignment (Section 6).

Eventually, since ED requires access to language model output token distributions over the vocabulary, this notably compromises the security of open-source models. Thus, we advocate for a critical reconsideration of the open accessibility of language models even those that have been safety-aligned.

2 Related Works

Safety alignment. Safety alignment enhances the helpfulness of language model responses to safe queries and prevents inappropriate responses to harmful queries (Bai et al., 2022; Touvron et al., 2023b). Most modern conversational language models are safety-aligned, either through deliberate safety tuning (Bai et al., 2022; Touvron et al., 2023b; Achiam et al., 2023; Zhou et al., 2023) or by learning from curated datasets that contain safety-related data (Jiang et al., 2023; Tunstall et al., 2023). However, our study suggests that these safety-aligned models can still be exploited to generate harmful responses without additional training.

Large language model attack. This work contributes to the field of LLM attacks, aiming to elicit harmful responses from large language models. We refer readers to the survey by Dong et al. (2024) for a comprehensive overview of LLM attack. While most studies focus on attacking language models within the input space through adversarial prompts (Zou et al., 2023; Shen et al., 2023; Liu et al., 2023; Li et al., 2023; Chao et al., 2023), our work targets the output space, manipulating the output token distributions of language models at inference time. Given that our method assumes access to the output token distributions of pre-trained models, which are already strong baselines for producing harm-

ful responses, our work should be seen as using safety-aligned models to further enhance the harmfulness of these pre-trained models. This approach surpasses merely jailbreaking the safety guardrails of aligned models.

Increasing LLM harmfulness via fine-tuning. This work also relates to the recent finding that LLM safety can downgrade with minimal fine-tuning (Qi et al., 2023; Zhao et al., 2024). However, our method does not involve actual fine-tuning or requires access to model weights; instead, we emulate fine-tuning through sampling (Mitchell et al., 2023; Liu et al., 2024), needing only access to language model output token distributions. Concurrently with our work, Zhao et al. (2024) propose a similar method that produces a harmful language model by combining the output distributions from different language models. Unlike their approach, which necessitates explicit fine-tuning to train a smaller unsafe model, our method is completely training-free, reusing off-the-shelf language models to produce harmful language models without additional training.

3 Preliminaries on Emulated Fine-Tuning (EFT)

Emulated disalignment builds on emulated fine-tuning (EFT) (Mitchell et al., 2023), which views the alignment of a language model π_{align} as a KL-constrained reward maximization problem:

$$\pi_{\text{align}} = \arg\max_{\pi} \mathbb{E}_{\mathbf{x}\sim p(\mathbf{x}),\mathbf{y}\sim\pi(\mathbf{y}|\mathbf{x})}[r_{\text{align}}(\mathbf{x},\mathbf{y}) - \mathbb{D}_{\text{KL}}(\pi(\mathbf{y}|\mathbf{x}) \,||\, \pi_{\text{base}}(\mathbf{y}|\mathbf{x}))], \tag{1}$$

where p is a distribution of queries $\mathbf{x}$, $\mathbf{y}$ is the language model response to a query $\mathbf{x}$, r_{align} is a reward function steering the language model to align with human intents, and $\mathbb{D}_{\text{KL}}(\pi(\mathbf{y}|\mathbf{x}) \,||\, \pi_{\text{base}}(\mathbf{y}|\mathbf{x}))$ is the KL-divergence from the pre-trained language model π_{base}. Conventionally, there is a hyperparameter β controlling the strength of the KL constraint, but in this paper, we omit writing β explicitly as it can always be subsumed into the reward by scaling it with β^{-1}. Prior work shows a mapping among π_{base}, π_{align} and r_{align} (Rafailov et al., 2024):

$$\pi_{\text{align}}(\mathbf{y}|\mathbf{x}) = \frac{1}{Z(\mathbf{x})}\pi_{\text{base}}(\mathbf{y}|\mathbf{x}) \exp\left(r_{\text{align}}(\mathbf{x},\mathbf{y})\right), \tag{2}$$

or equivalently,

$$r_{\text{align}}(\mathbf{x},\mathbf{y}) = \log \frac{\pi_{\text{align}}(\mathbf{y}|\mathbf{x})}{\pi_{\text{base}}(\mathbf{y}|\mathbf{x})} + \log Z(\mathbf{x}), \tag{3}$$

where $Z(\mathbf{x}) = \sum_{\mathbf{y}} \pi_{\text{base}}(\mathbf{y}|\mathbf{x}) \exp(r_{\text{align}}(\mathbf{x},\mathbf{y}))$ is the partition function. This mapping not only expresses a duality between language models and reward functions but also has an important practical implication: Eq. 3 enables "reverse-engineering" the proprietary reward functions that produce the language models, assuming access to the language model output token distributions. For example, Mitchell et al. (2023) use $\log \pi_{\text{Llama-2-7b-chat}}(\mathbf{y}|\mathbf{x}) - \log \pi_{\text{Llama-2-7b}}(\mathbf{y}|\mathbf{x})$ (Touvron et al., 2023b) as a proxy for the closed-source reward function to guide the decoding of a larger base model, thereby emulating a larger aligned model.

4 Emulated Disalignment (ED)

Given the reverse-engineered reward function $r_{\text{align}}(\mathbf{x},\mathbf{y}) = \log \pi_{\text{align}}(\mathbf{y}|\mathbf{x}) - \log \pi_{\text{base}}(\mathbf{y}|\mathbf{x})$ from the $(\pi_{\text{align}}, \pi_{\text{base}})$ pair (Eq. 3), emulated disalignment (ED) demonstrates that this reward function can be exploited to create a harmful language model, especially when π_{align} is **safety-aligned**. Specifically, ED incorporates the following three insights into one training-free attack method: **(1)** If π_{align} is safety-aligned, the reverse-engineered reward function r_{align} is a safety reward function penalizing harmful responses; **(2)** Adversarially fine-tuning the pre-trained language model π_{base} to minimize r_{align} leads to a harmful language model; **(3)** Crucially, the result of such adversarial fine-tuning can be efficiently emulated by pure sampling, eliminating the need for additional training.

4.1 Deriving the ED Sampling Distribution

Here, we detail the three insights mentioned above and derive the ED sampling distribution.

(1) Reverse-engineering a safety reward function from the pre-trained and safety-aligned language model pair. Theoretically, the duality between reward functions and language models (Eq. 3) suggests that the reverse-engineered reward function r_{align} should reflect the fine-tuning principles used to obtain the safety-aligned model ($\pi_{\text{base}} \rightarrow \pi_{\text{align}}$). Given that safety alignment aims to reduce the harmfulness of responses to harmful queries and increase the helpfulness of responses to safe queries (Bai et al., 2022; Touvron et al., 2023b), r_{align} should be a safety reward function that *penalizes harmful responses to harmful queries* and encourages helpful responses to safe queries. We focus on the former, examining how this reward supports adversarial purposes.

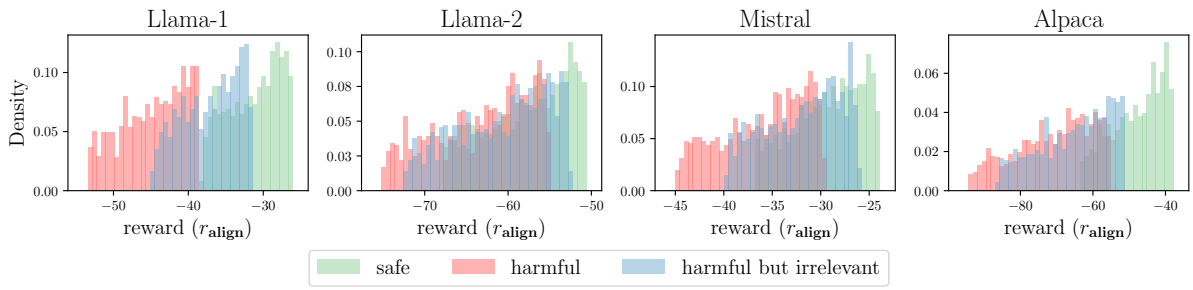


Figure 3: Distributions (bottom 15%) of reverse-engineered reward function $r_{\text{align}}(\mathbf{x}, \mathbf{y}) = \log \pi_{\text{align}}(\mathbf{y}|\mathbf{x}) - \log \pi_{\text{base}}(\mathbf{y}|\mathbf{x})$ on different types of responses $\mathbf{y}$ to the same harmful queries $\mathbf{x}$ from the HH ("harmless-base") dataset (Bai et al., 2022). For each query, safe indicates the preferred response to this query, harmful indicates the dispreferred response to this query, and harmful but irrelevant indicates the dispreferred response to a random query. These plots suggest that reversed-engineered r_{align} encourage safe responses and **penalize harmful responses.**

Empirically, we verify that r_{align} is a safety reward by analyzing its distributions on different types of responses to the same harmful queries. Figure 3 shows r_{align}, reverse-engineered from different model families (model specifications in Section 5), consistently assign higher rewards to safe responses and lower rewards to harmful responses.

(2) Disalignment: Minimizing this safety reward leads to a harmful language model. Given a safety reward function r_{align} penalizing harmful responses to harmful queries, the next step to create a harmful language model that supports harmful queries is to minimize r_{align} (as opposed to maximization in Eq. 1; note the negative sign below):

$$\pi_{\text{disalign}} = \arg\max_{\pi} \mathbb{E}_{\mathbf{x}\sim p(\mathbf{x}), \mathbf{y}\sim\pi(\mathbf{y}|\mathbf{x})}[-\alpha r_{\text{align}}(\mathbf{x}, \mathbf{y}) - \mathbb{D}_{\text{KL}}(\pi(\mathbf{y}|\mathbf{x}) \,||\, \pi_{\text{base}}(\mathbf{y}|\mathbf{x}))], \tag{4}$$

where $\alpha > 0$ is a positive hyperparameter controlling the trade-off between minimizing the reward and the KL constraint. We define this reward minimization problem as *disalignment* as it steers the language models in the exact opposite direction of alignment. Usually, solving Eq. 4 requires resource-intensive training (e.g., reinforcement learning). However, we will demonstrate that the result of this optimization can be emulated efficiently without additional training, resulting in a high-stakes adversarial attack framework.

(3) Emulating disalignment through pure sampling. To obtain the harmful language model π_{disalign} from Eq. 4, instead of optimizing directly with reinforcement learning, combining Eq. 2 and Eq. 3 enables the result of disalignment to be expressed in a closed form without training:

$$\begin{aligned} &\pi_{\text{disalign}}(\mathbf{y}|\mathbf{x}) \\ &\propto \pi_{\text{base}}(\mathbf{y}|\mathbf{x}) \exp\left(-\alpha r_{\text{align}}(\mathbf{x}, \mathbf{y})\right) && \text{(Eq. 2)} \\ &\propto \pi_{\text{base}}(\mathbf{y}|\mathbf{x}) \exp\left(-\alpha \log \frac{\pi_{\text{align}}(\mathbf{y}|\mathbf{x})}{\pi_{\text{base}}(\mathbf{y}|\mathbf{x})}\right) && \text{(Eq. 3)} \\ &= \frac{\pi_{\text{base}}(\mathbf{y}|\mathbf{x})^{\alpha+1}}{\pi_{\text{align}}(\mathbf{y}|\mathbf{x})^{\alpha}}. && (5) \end{aligned}$$

These derivations are exact, but this sequence-level distribution is incompatible with the autoregressive nature of modern generative large language models. To address this, we follow the per-token approximation trick from EFT (Mitchell et al., 2023) and sample from the following autoregressive distribution to approximate π_{disalign}:

$$\boxed{\pi_{\text{emulated-disalign}}(\mathrm{y}_t|\mathbf{x}, \mathbf{y}_{<t}) \propto \frac{\pi_{\text{base}}(\mathrm{y}_t|\mathbf{x}, \mathbf{y}_{<t})^{\alpha+1}}{\pi_{\text{align}}(\mathrm{y}_t|\mathbf{x}, \mathbf{y}_{<t})^{\alpha}},} \tag{6}$$

where $\mathbf{y}_{<t}$ denotes all response tokens up to the $(t-1)$th token. Although this per-token approximation to the sequence-level distribution has a loosely bounded regret (Haarnoja et al., 2018), it effectively circumvent the cumbersome fine-tuning process and demonstrates strong empirical performance (more details in Section 5, 6).

ED: Putting it all together. Essentially, Eq. 6 is all we need for reversing safety alignment: at the core of ED is a straightforward sampling distribution that combines the output token distributions of safety-aligned and pre-trained language models, which we justify within the framework of reward minimization. We call this method *emulated disalignment (ED)* as it emulates the outcome of disalignment (Eq. 4) without additional training and